

Assessing Wildfire Impacts on Terrestrial Vertebrate Biodiversity in Los Angeles Using MaxEnt and Remote Sensing

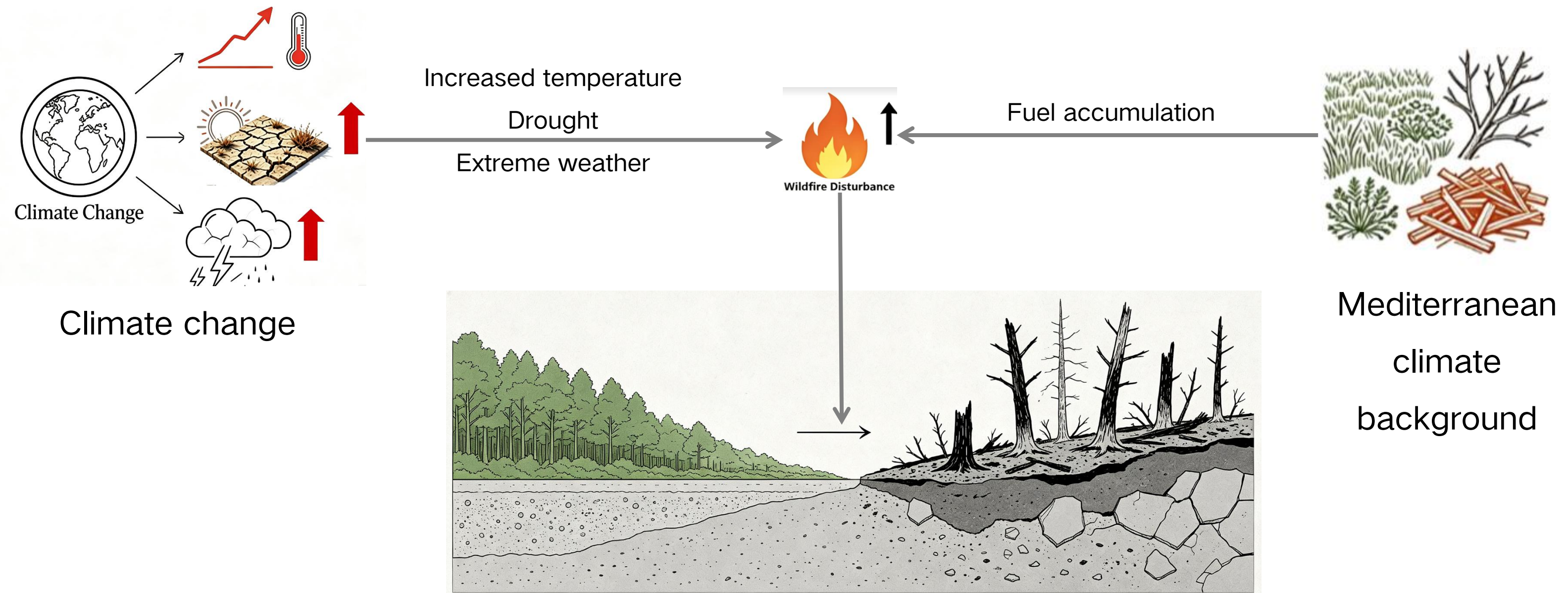
Junyu Deng¹, Sisi Chen², Zengrui Tian², Zhijun You³

¹Shanghai Guanghua Cambridge International School,

²Zhejiang University, ³Beijing Forestry University

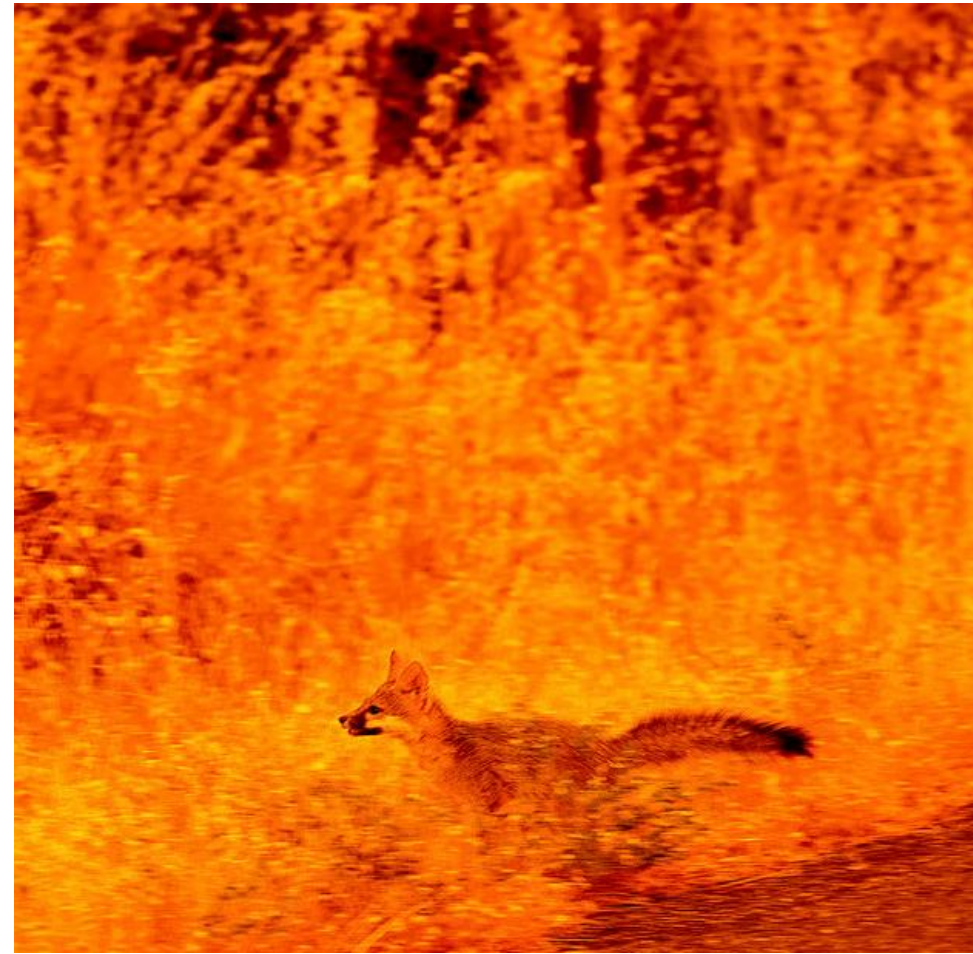


Background



- **Global warming** has intensified the frequency and severity of extreme wildfires
- Southern California's **Mediterranean climate**, characterized by wet winters and dry summers, promotes rapid biomass accumulation followed by abundant summer fuels

Background



- **January 2025, Los Angeles experienced two major wildfires—the Eaton Fire and the Palisades Fire—ranked as the second and third most destructive in California’s history**
- **the socio-economic impacts of these fires have been widely reported, *their effects on suitable habitats and biodiversity remain poorly understood***

Aims and Scope

Research Focus

Major wildfires in Los Angeles affect the availability of suitable habitats.

Habitat Suitability Change

Q1: Fire extent and intensity

Q2: Habitat suitability change

Q3: Species tolerance to fire

Fire Severity Mapping

Burned Areas

Burned Intensity t

Habitat Suitability Change

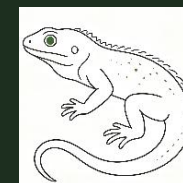
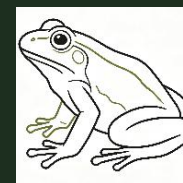
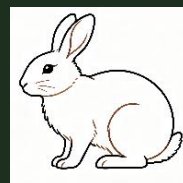


Before Fire



After Fire

Species Response



Escape or Adaptation?

Study area

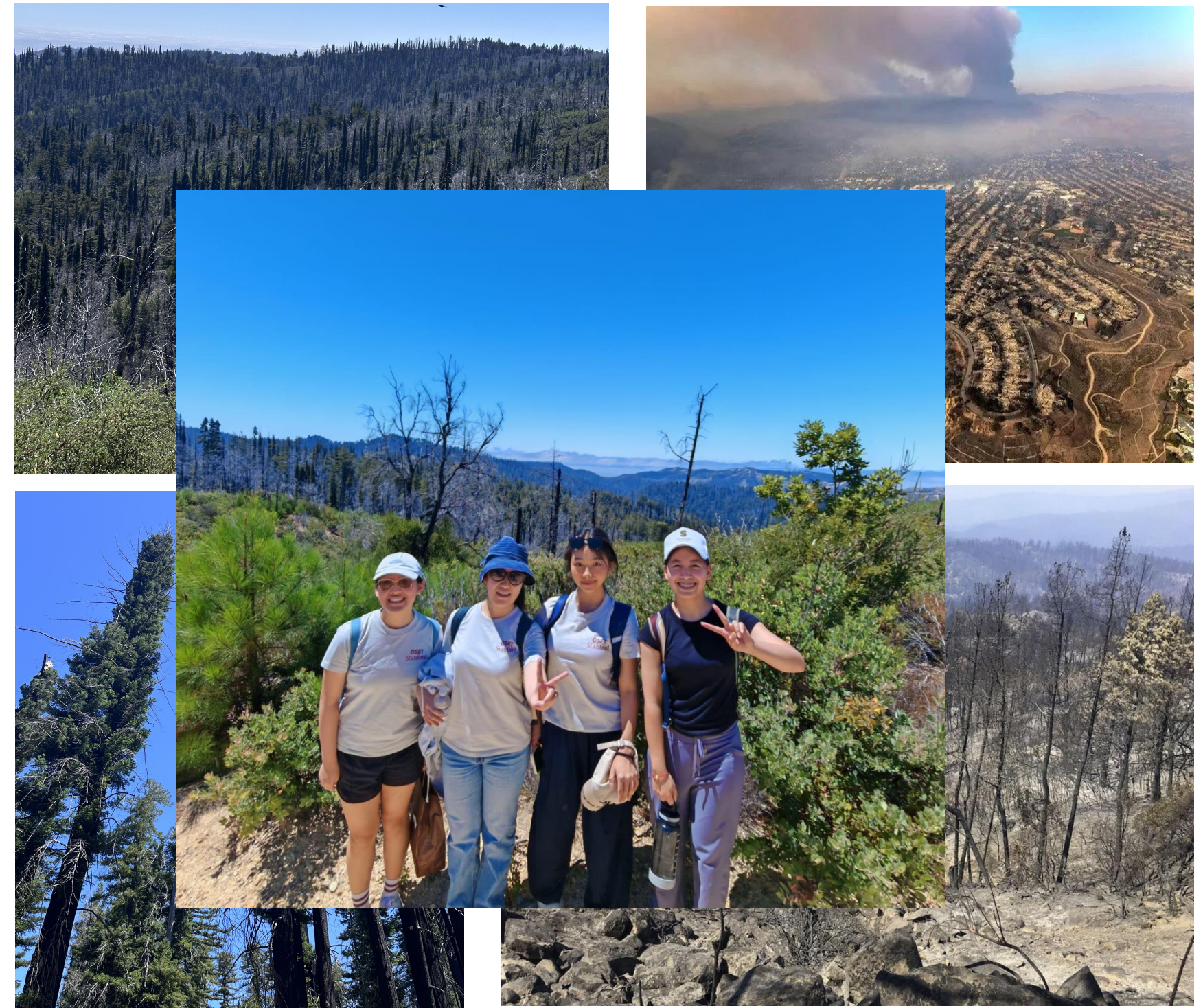
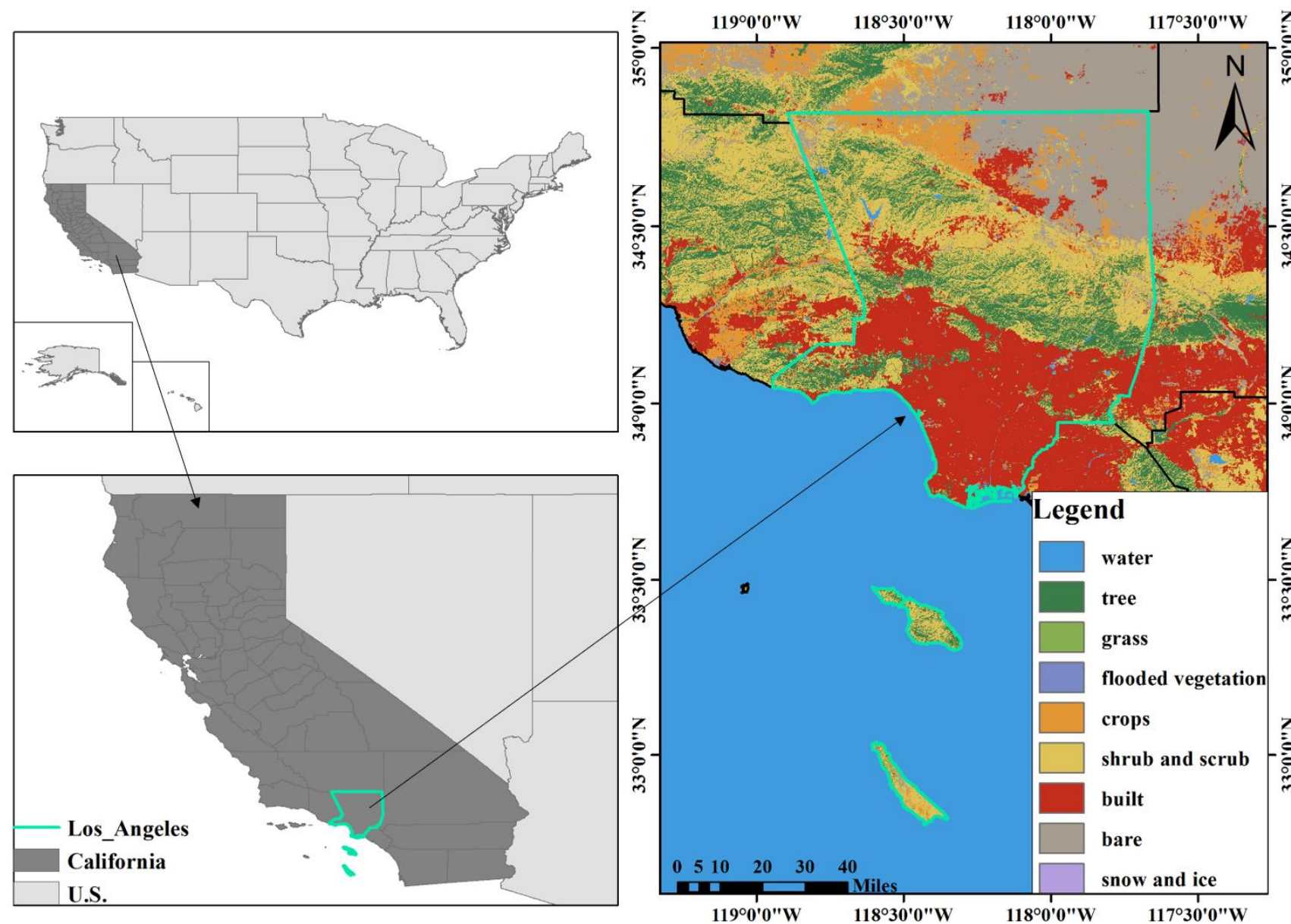
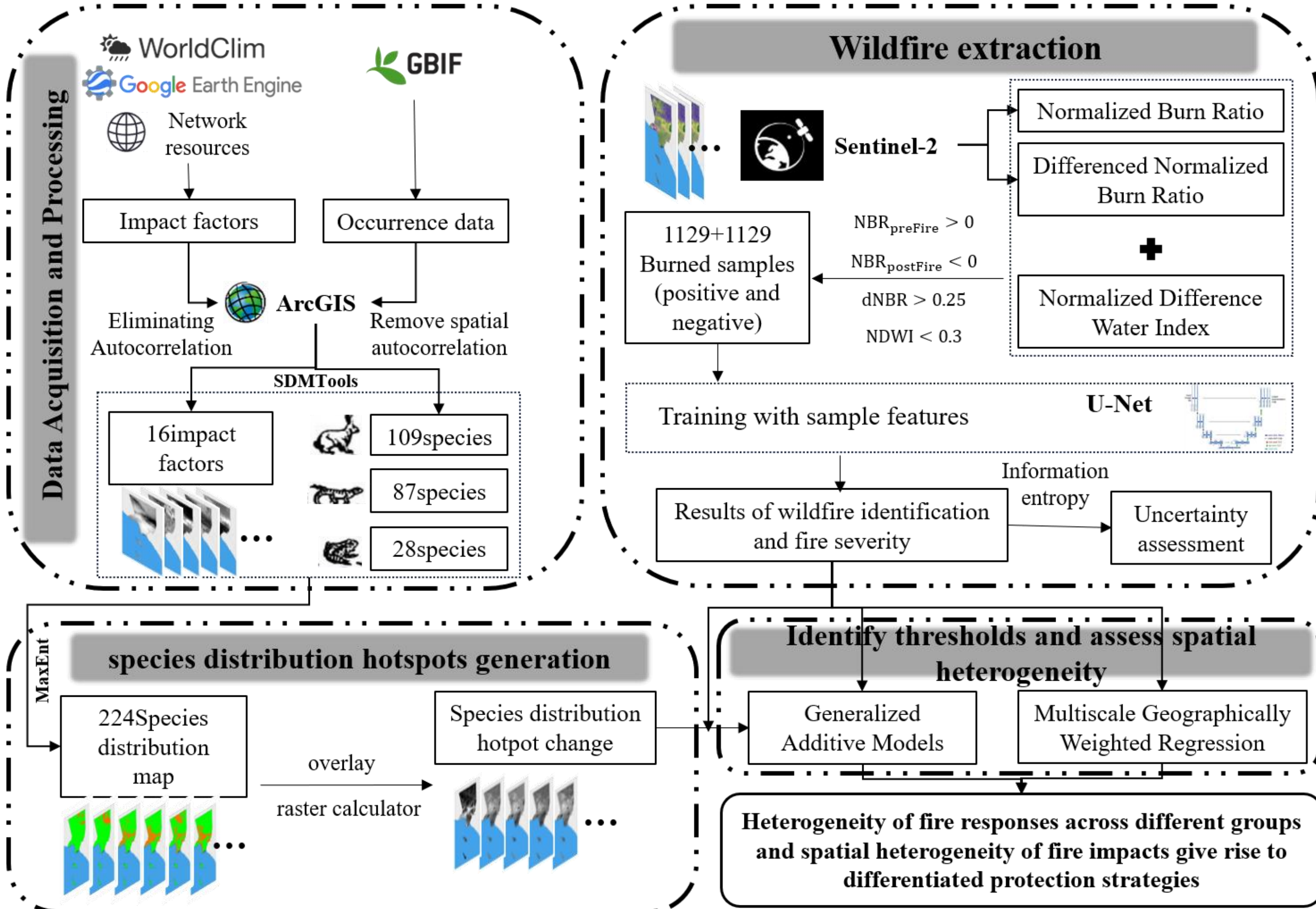


Figure. Overview of the study area. Left: location of the study area; Right: land use and land cover within the study area

- Given Los Angeles's unique land-use patterns, rich biodiversity and long history of wildfires, it makes sense to select **Los Angeles** as the study area to investigate the species' responses to fire

Methodology



Method-Fire identification

Data Source:

Sentinel-2 imagery (2025)

Spatial resolution: 10 m

Indices:

$$\mathbf{NBR} = (\rho_{\mathbf{NIR}} - \rho_{\mathbf{SWIR2}}) / (\rho_{\mathbf{NIR}} + \rho_{\mathbf{SWIR2}})##$$

$$\mathbf{N} = (\rho_{\mathbf{Green}} - \rho_{\mathbf{NIR}}) / (\rho_{\mathbf{Green}} + \rho_{\mathbf{NIR}})#$$

$$\mathbf{dNBR} = \mathbf{NBR}_{\mathbf{preFire}} - \mathbf{NBR}_{\mathbf{postFire}}#$$

Burn Severity Level

dNBR Range

High growth

< -0.25

Low growth

-0.25~-0.1

Unburnt

-0.1~0.1

Low

0.1~0.27

Moderate-low

0.27~0.44

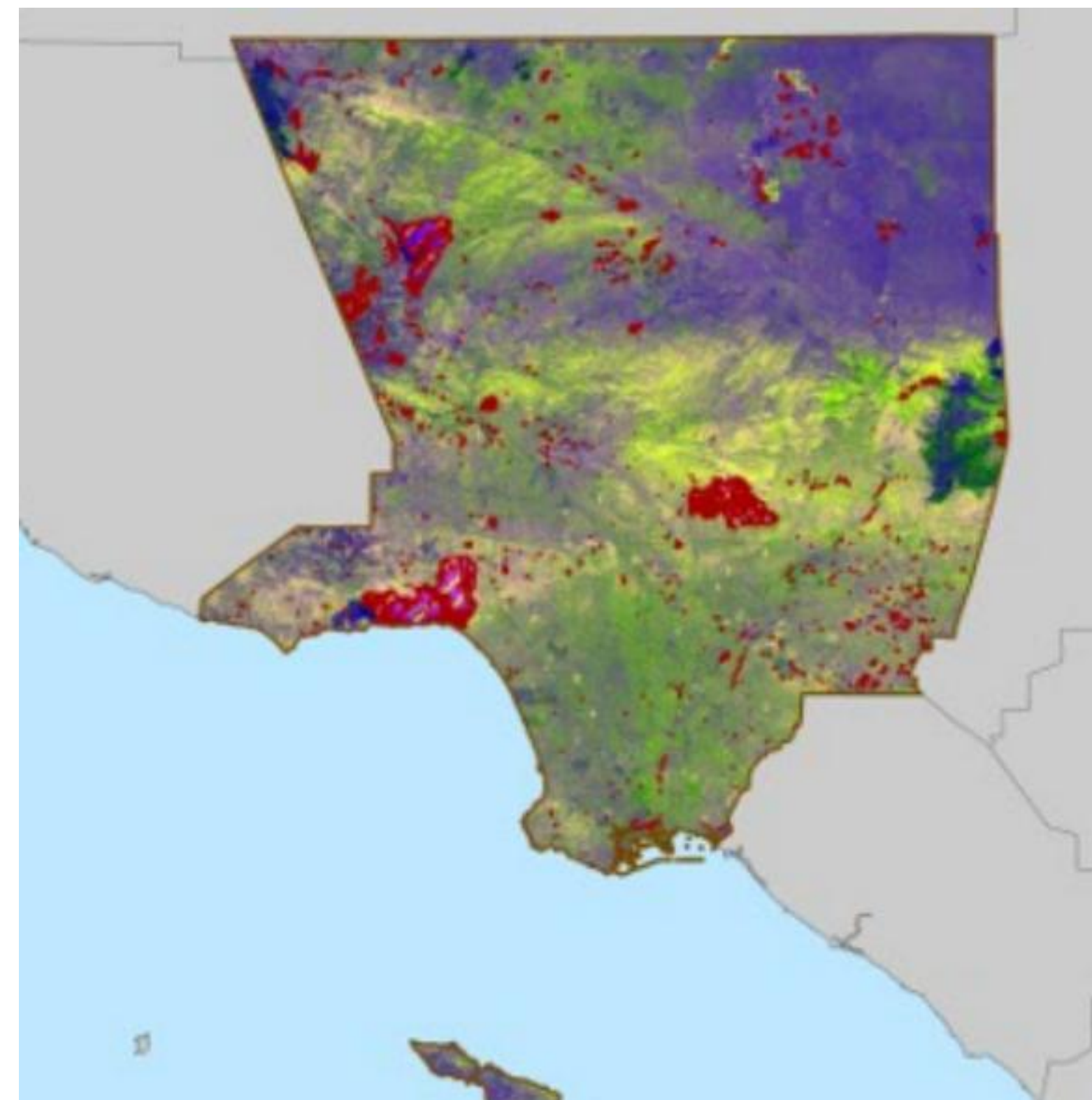
Moderate-high

0.44~0.66

High

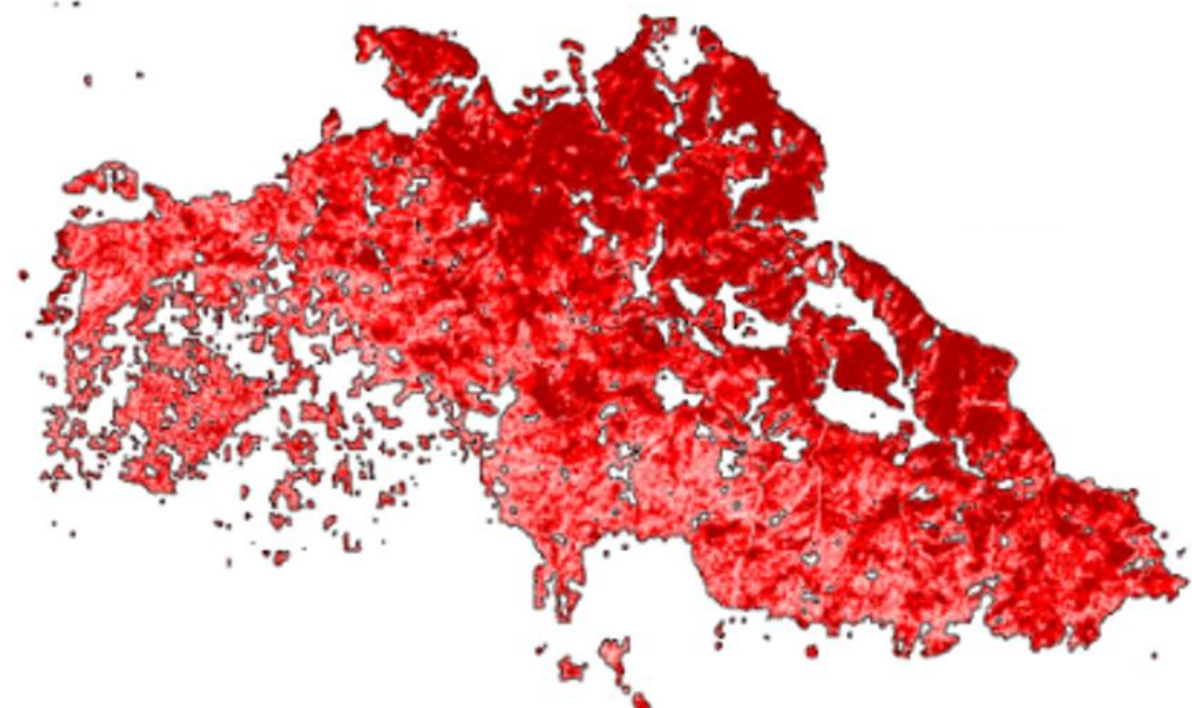
> 0.66

Fire severity classification threshold based on dNBR

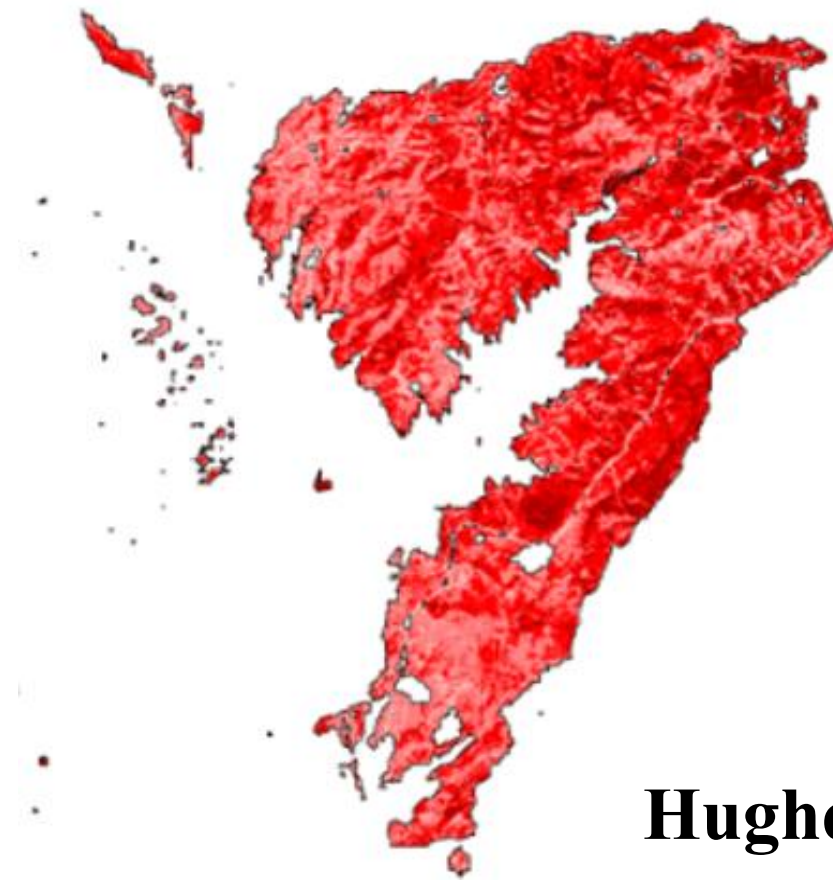


Imaginary example

Method-Fire identification



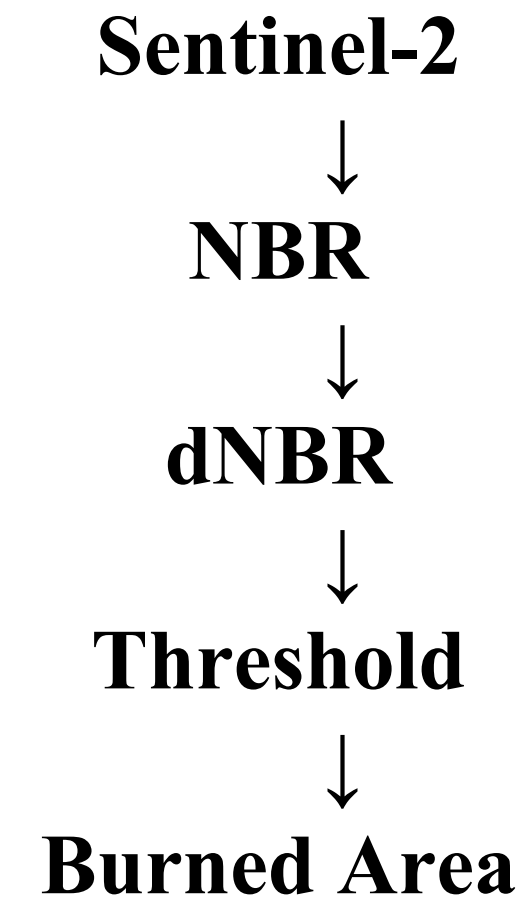
Eaton



Hughes



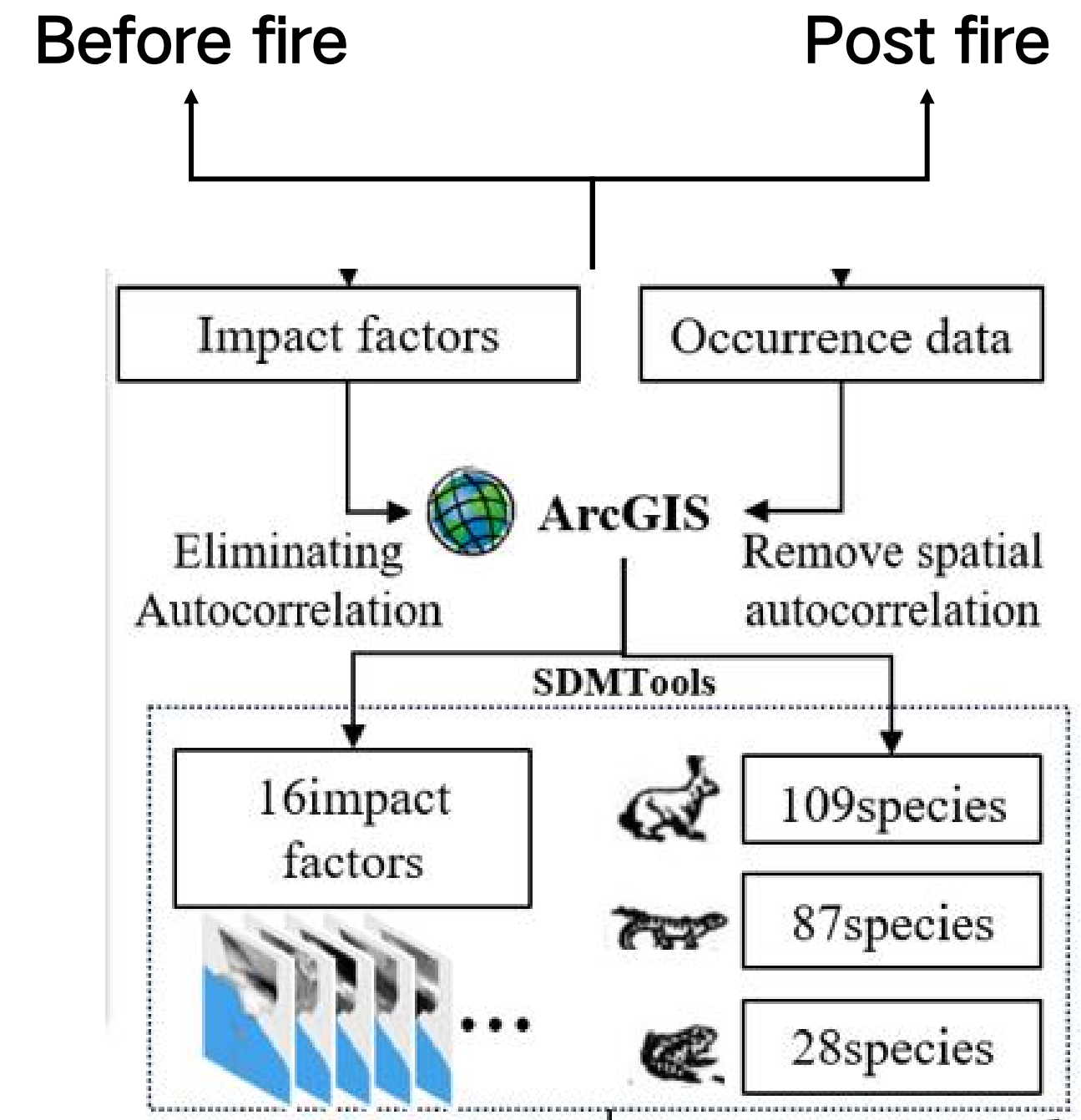
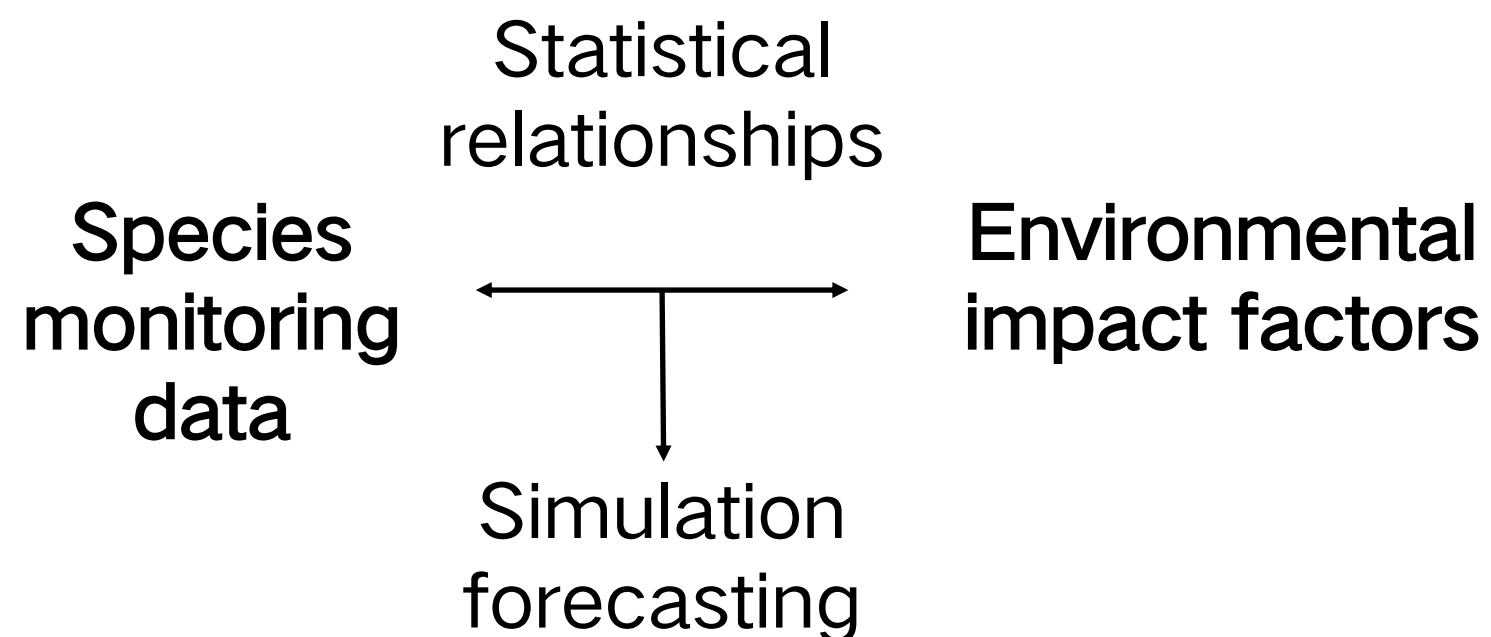
Palisades



Burned areas extracted from Sentinel-2 imagery using the **dNBR-based method** for three representative wildfire cases. **Red pixels** indicate the detected burned areas.

Method-Habitat change

Species Distribution Models (**SDMs**), commonly referred to as niche modelling, it is currently the primary tool used to derive spatially explicit **predictions of species' environmental suitability**, and is the **most widely used** modelling tool for predicting the impacts of global change on biodiversity.





Method-Impact factors

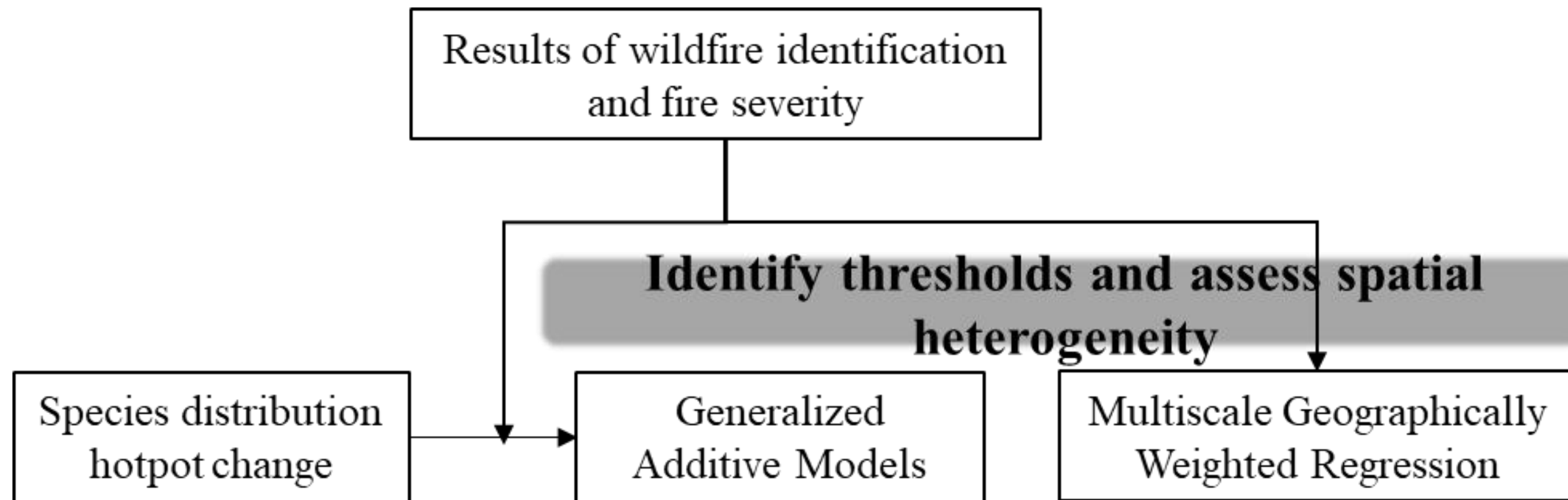
Thirty variables were selected across three dimensions: climate, environment, and human disturbance, based on regional characteristics and previous habitat studies.



30 impact factors
SDMTtools
 $|r| \geq 0.8$
16 impact factors

variable cate- gory	variable name	resolution	source
bioclimatic variables	Annual mean temperature	~1km	https://worldclim.org/
	Isothermality		
	Maximum temperature of warmest month		
	Mean temperature of driest quarter		
	Annual precipitation		
	Precipitation seasonality		
	Precipitation of driest quarter		
Environmental Variables	Aspect	30m	https://opentopography.org/
	Slope		
	FVC	100m	Based on the Landsat Collection 2 Level-2 Surface Reflectance prod- uct, see Appendix A.2 for the index calculation. Based on the HydroSHEDS dataset, https://www.hydrosheds.org/prod- ucts/hydrorivers, see Appendix A.2 for the index processing. Dynamic World V1, the data we used is from the year 2024 and 2025, based on Google Earth En- gine. generated from climate data, see Appendix A.2 for the index pro- cessing.
	NDWI		
	TWI		
	LULC		
	AI		
human disturb- ance variables	Distance to water	500m	Road and surface water data are sourced from OpenStreetMap (OSM): https://download.geofab- rik.de/north-america/us/califor- nia/socal.html, see Appendix A.2 for details on data processing.
	Distance to roads		

Method-Species response



Generalized Additive Model

(GAM)

$$\mathbf{g}(\mathbf{E}(\mathbf{Y})) = \beta_0 + \sum_{j=1}^p \mathbf{s}_j(\mathbf{X}_j) \#$$

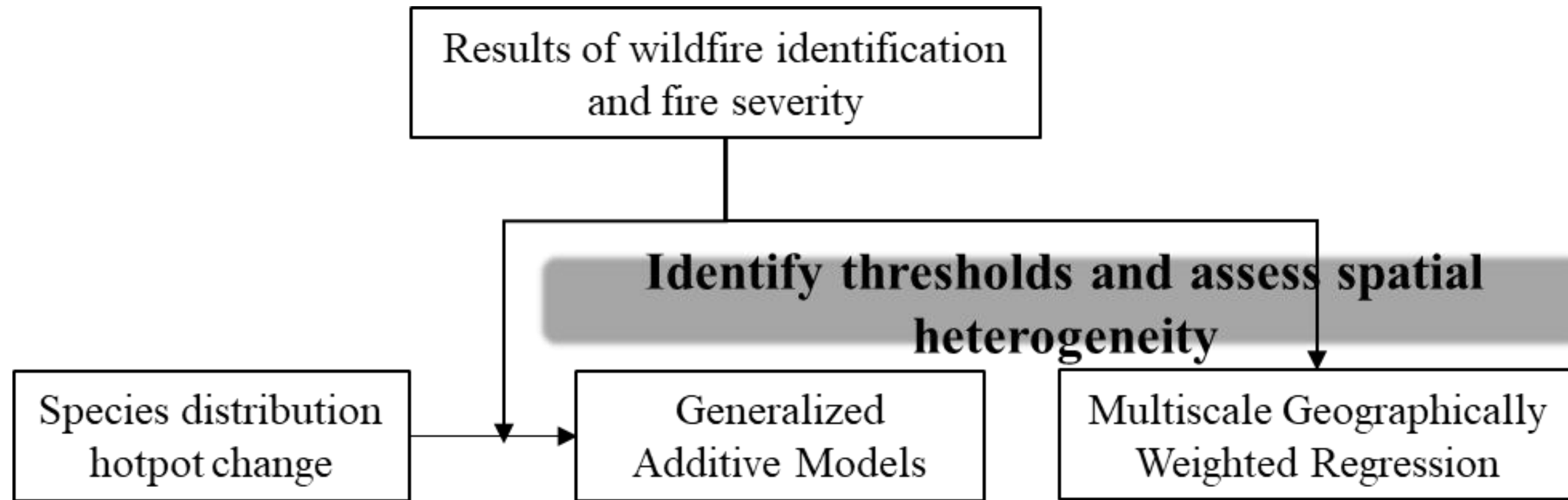
Features

- Models nonlinear relationships
- Uses smoothing splines
- Identity link function

Variables

- Response (Y): Species response / habitat suitability
- Predictor (X): Burn severity (dNBR) + environmental variables

Method-Species response



Multiscale Geographically Weighted Regression (MGWR)

$$Y_i = \beta_0(u_i, v_i) + \sum_{j=1}^k \beta_{bwj}(u_i, v_i) X_{ij} + \epsilon_i \#$$

Features

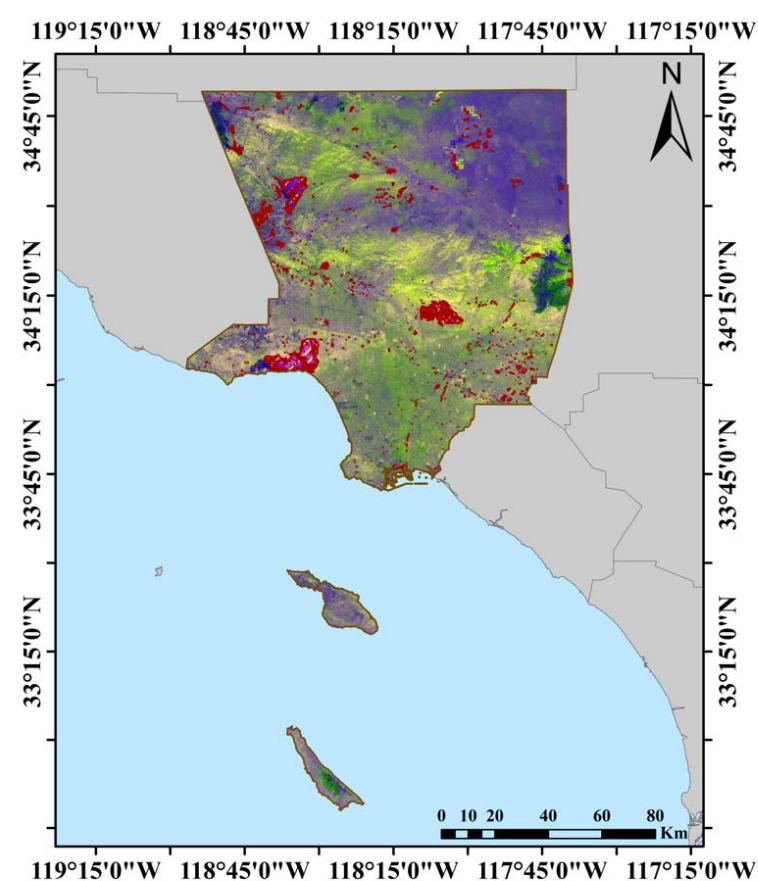
- Variable-specific spatial bandwidths
- Captures multiscale spatial relationships
- Estimates local effects of wildfire severity

Variables

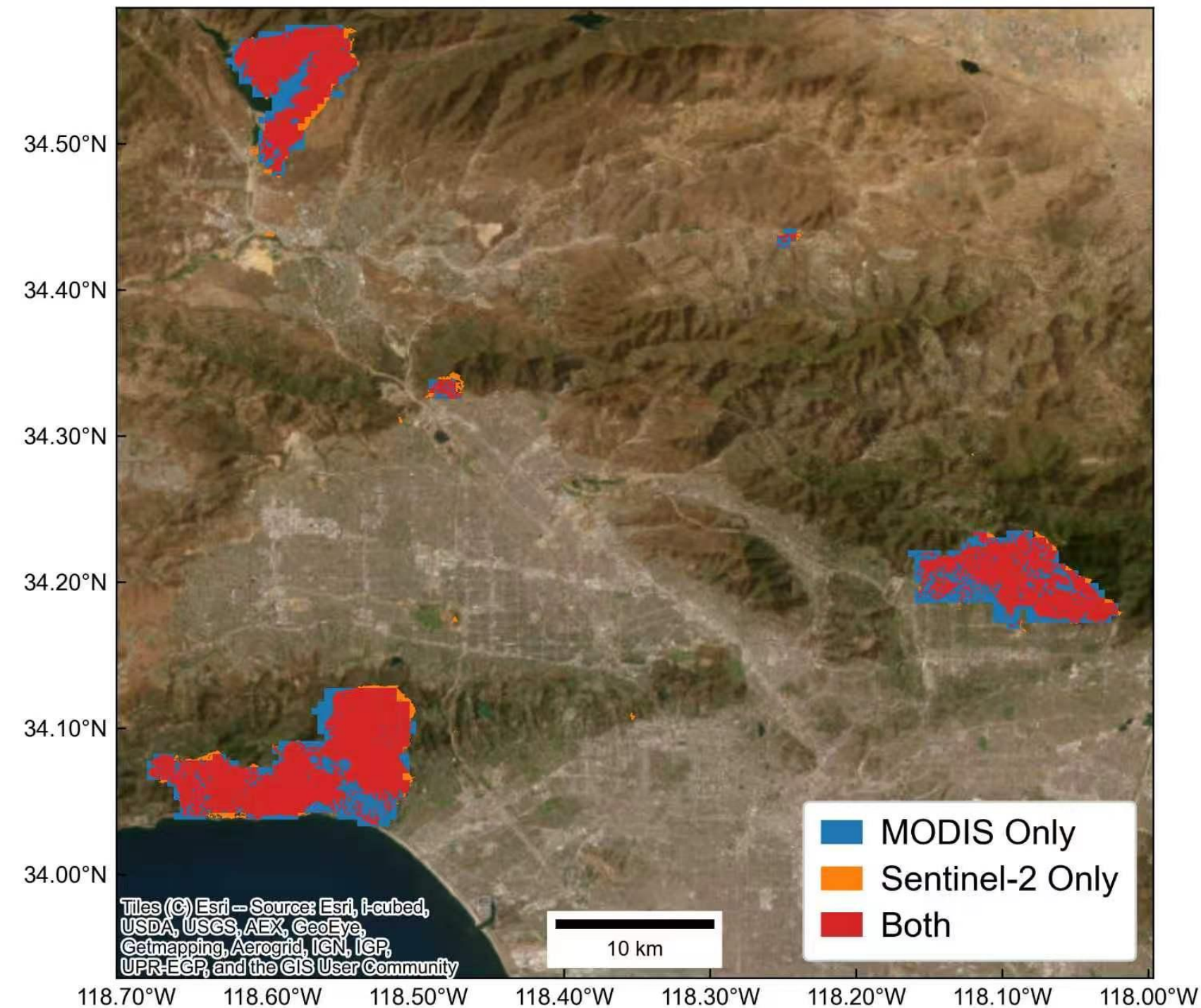
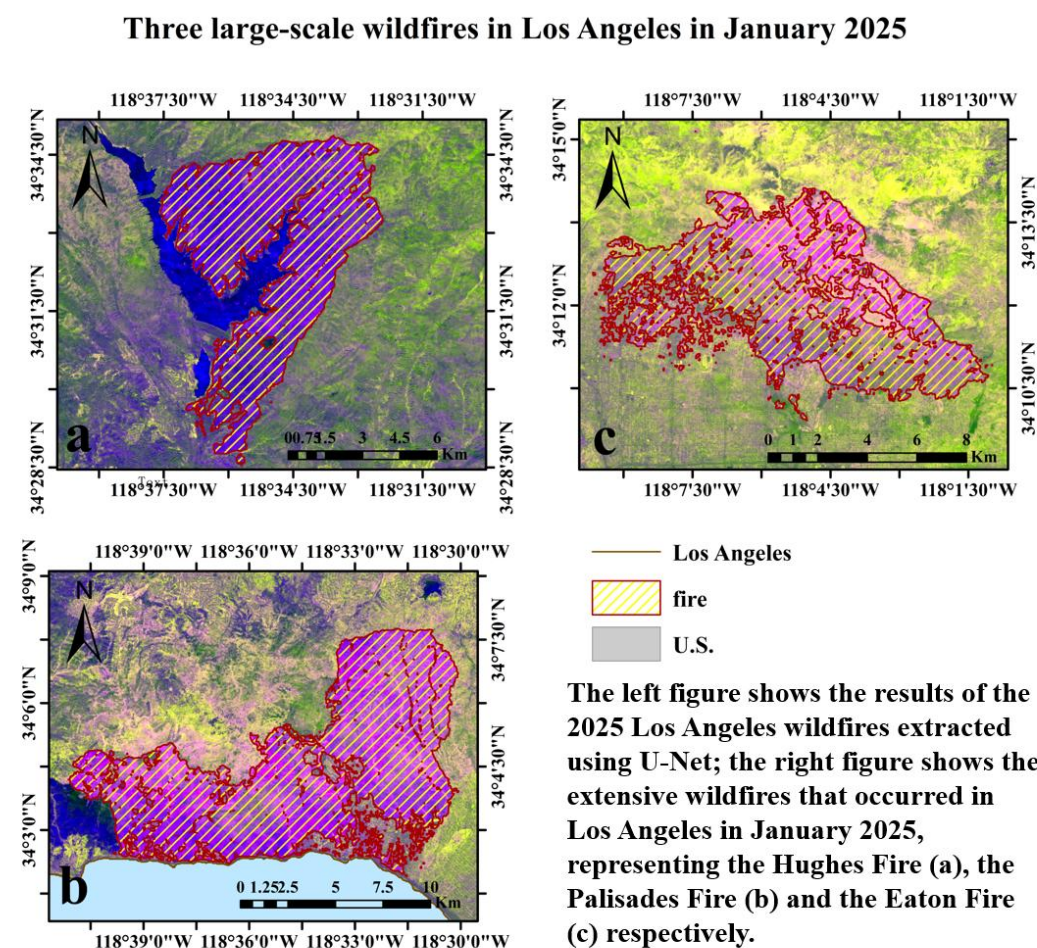
- Response: Biodiversity change (Δ Hotspot)
- Predictors: dNBR + topographic variables

Result-Fire extent and intensity

- The extracted burned areas captured the overall fire patterns with high spatial detail.
- The total burned area was **213.52 km²**.
- Palisades (82.48 km²), Eaton (41.21 km²), and Hughes (40.48 km²) were the three largest fires

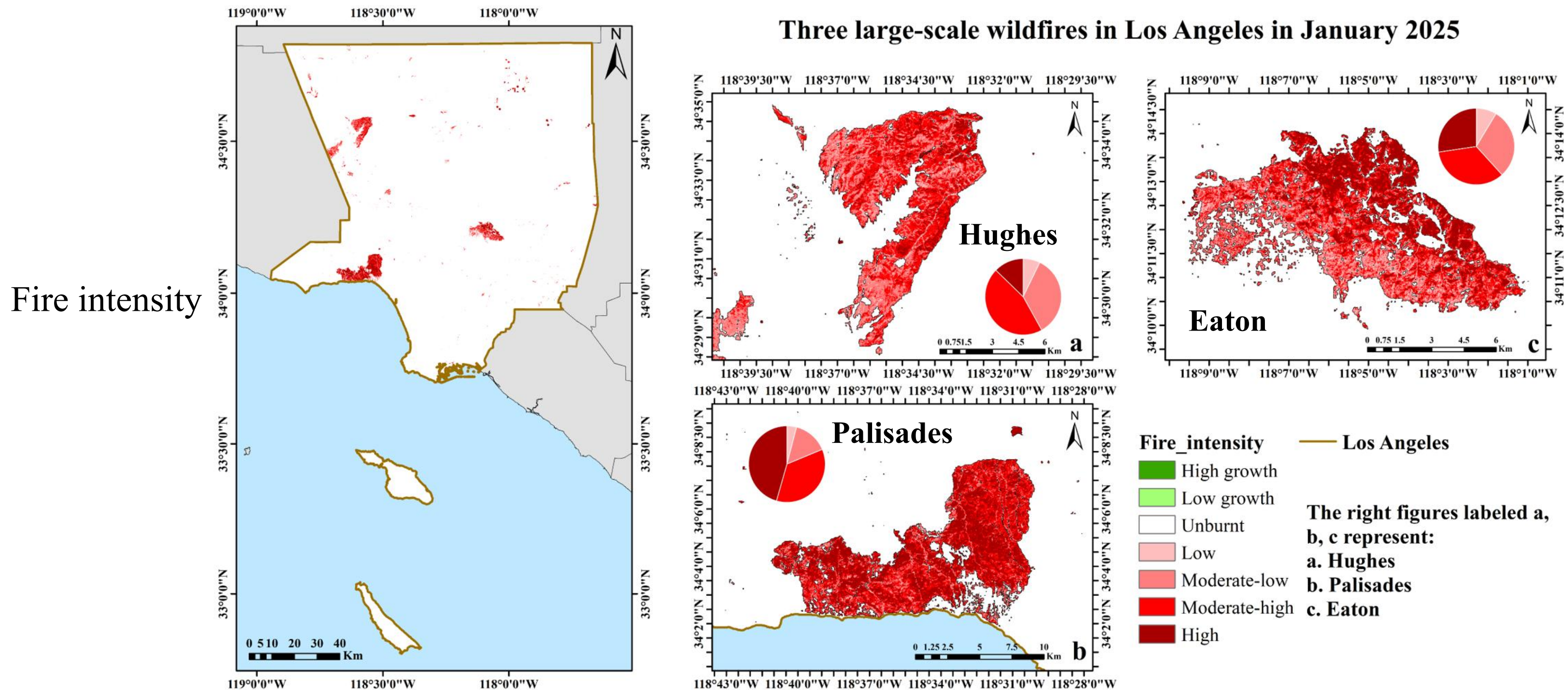


Fire extent



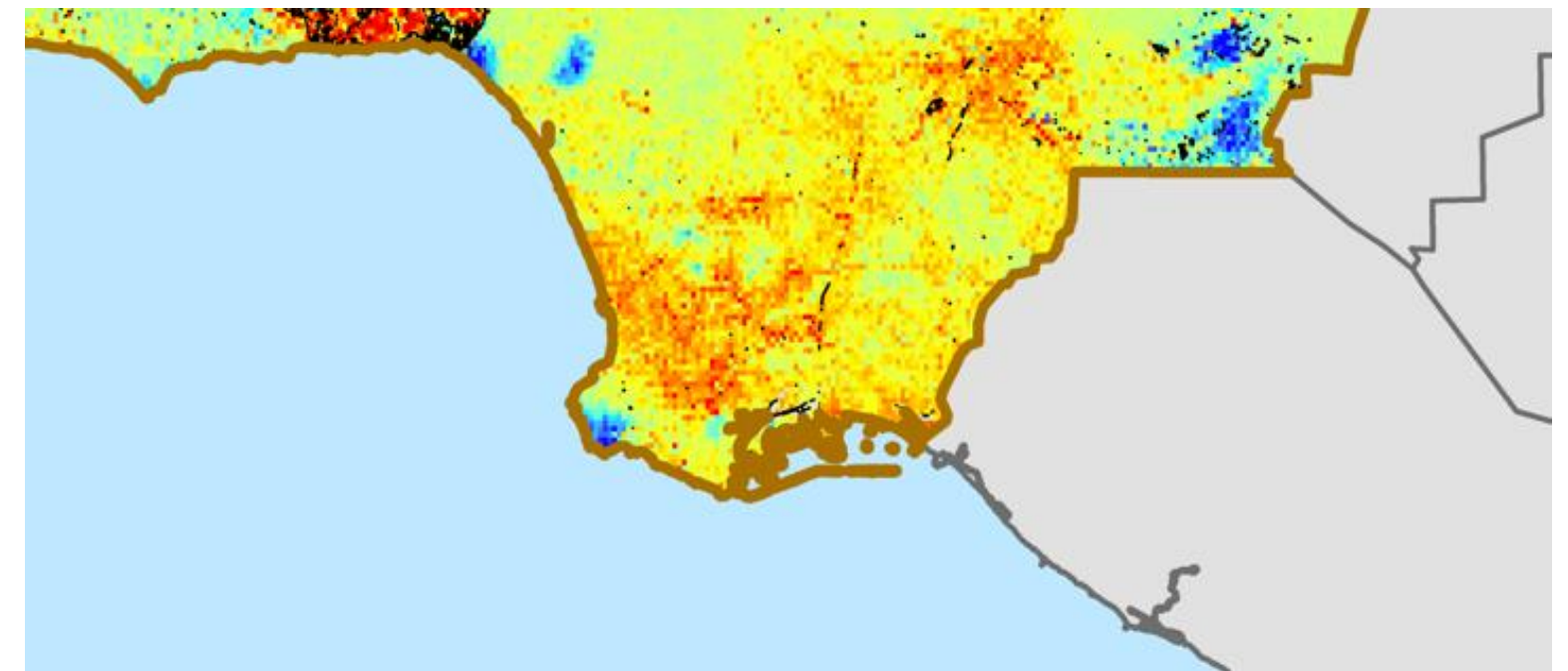
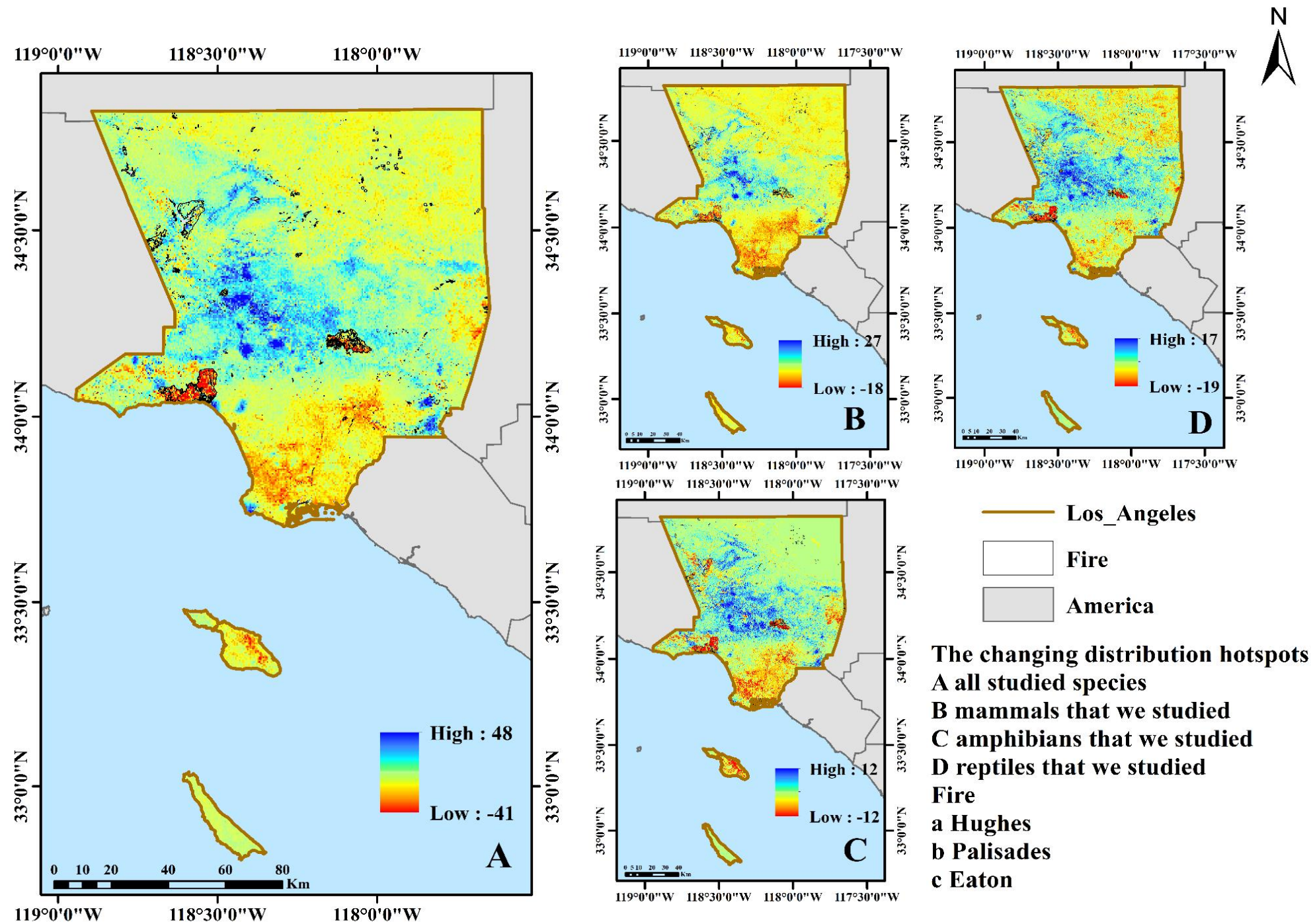
Result-Fire extent and intensity

- The burned areas in Los Angeles exhibited **relatively high burn severity**
- Among the three major wildfires with the largest affected areas, **the Palisades Fire** displayed a predominantly large proportion of high-severity burn areas, accounting for **45.58%** of its total extent, followed by the Eaton Fire at 27.35%



Result-Habitat change

- ❑ **56.1%** of the study area experienced biodiversity loss ($\Delta\text{Hotpot} < 0$).
- ❑ The most severe biodiversity loss:core zones of the **Palisades fire**
- ❑ Amphibians were most vulnerable, with **71.0%** of the study area showing species decline, followed by reptiles (65.2%) and mammals (55.0%).

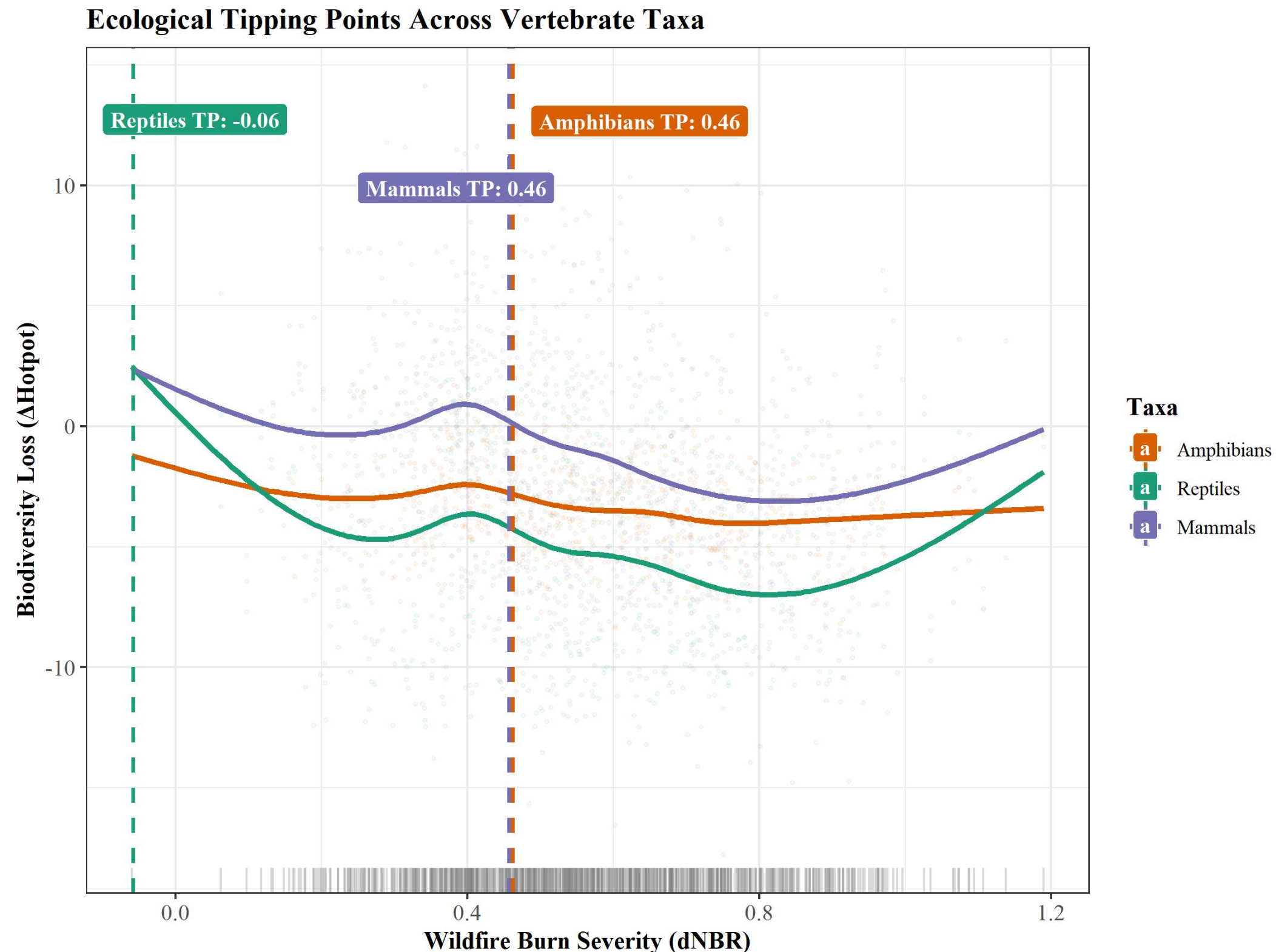


- ❑ areas that locate far away from the wildfire also experienced some species' decline

Result-Species response

Ecological Tipping Points

- ❑ Reptiles: ecological tipping point at **dNBR = -0.06**, within the Unburnt category.
- ❑ Amphibians & mammals: tipping point at **dNBR = 0.46**, corresponding to Moderate-high burn severity.



Result-Species response

MGWR Results

□ Hughes Fire

No significant wildfire effect

□ Palisades Fire

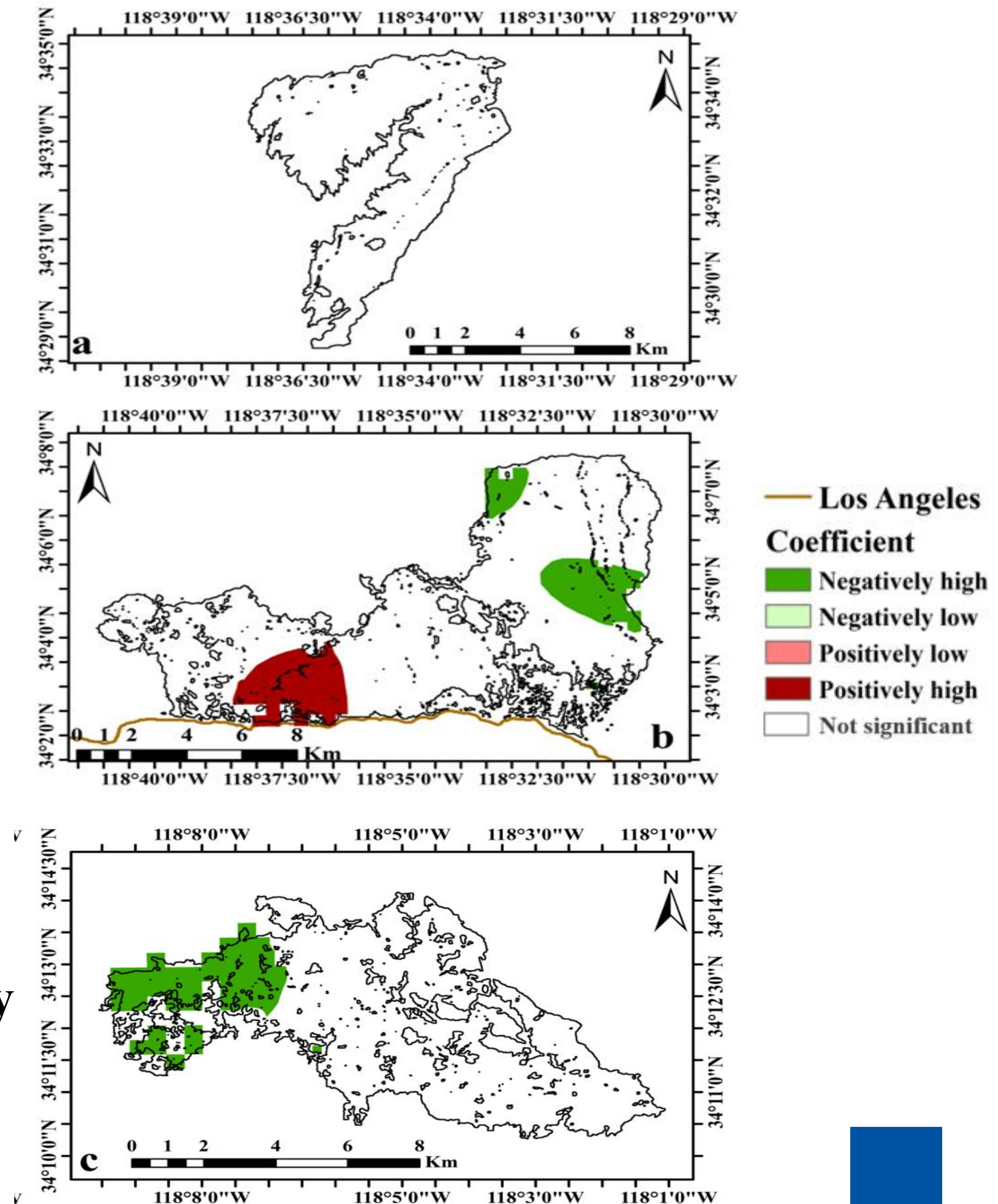
Spatially heterogeneous responses

South: Positive relationship

North & East: Negative relationship

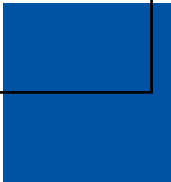
□ Eaton Fire

Higher burn severity → Greater biodiversity





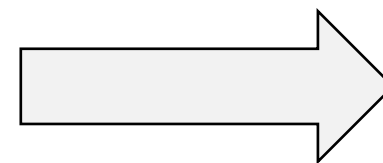
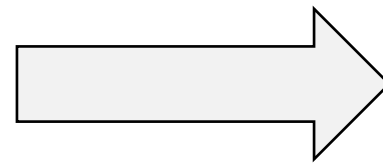
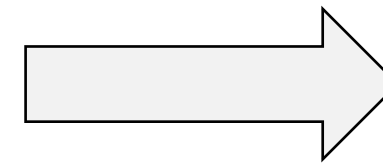
Post-fire recovery strategies

- **Spatial conservation planning:** Proactive ecological restoration must implement a **complementarity-based spatial optimization framework**
 - **For severely impacted amphibians and reptiles:** prioritize the physical stabilization and creation of micro-refugia within low-intensity burn islands.
 - **For mammals:** designate **dynamic, non-intervention buffer zones** along fire boundaries where opportunistic species congregate. Restrict human disturbance and salvage logging to preserve migration corridors and post-fire food resources.
- 

Methodological Limitations and Future Directions

Limitation

- **GBIF spatial sampling bias may reduce prediction accuracy in remote areas.**
- **The 10 m resolution may underestimate species survival during wildfires.**
- **Lack of post-fire field validation**
- **Assumed stable post-fire bioclimatic conditions, causing uncertainty in predictions.**



Future direction

- **Integrating multiple citizen science datasets for cross-validation**
- **Use high-resolution satellite or UAV/LiDAR data to map microhabitats and microrefugia.**
- **Establish an integrated air–space–ground monitoring network for post-fire field validation.**

Thank you for your patience
Feel free to ask!